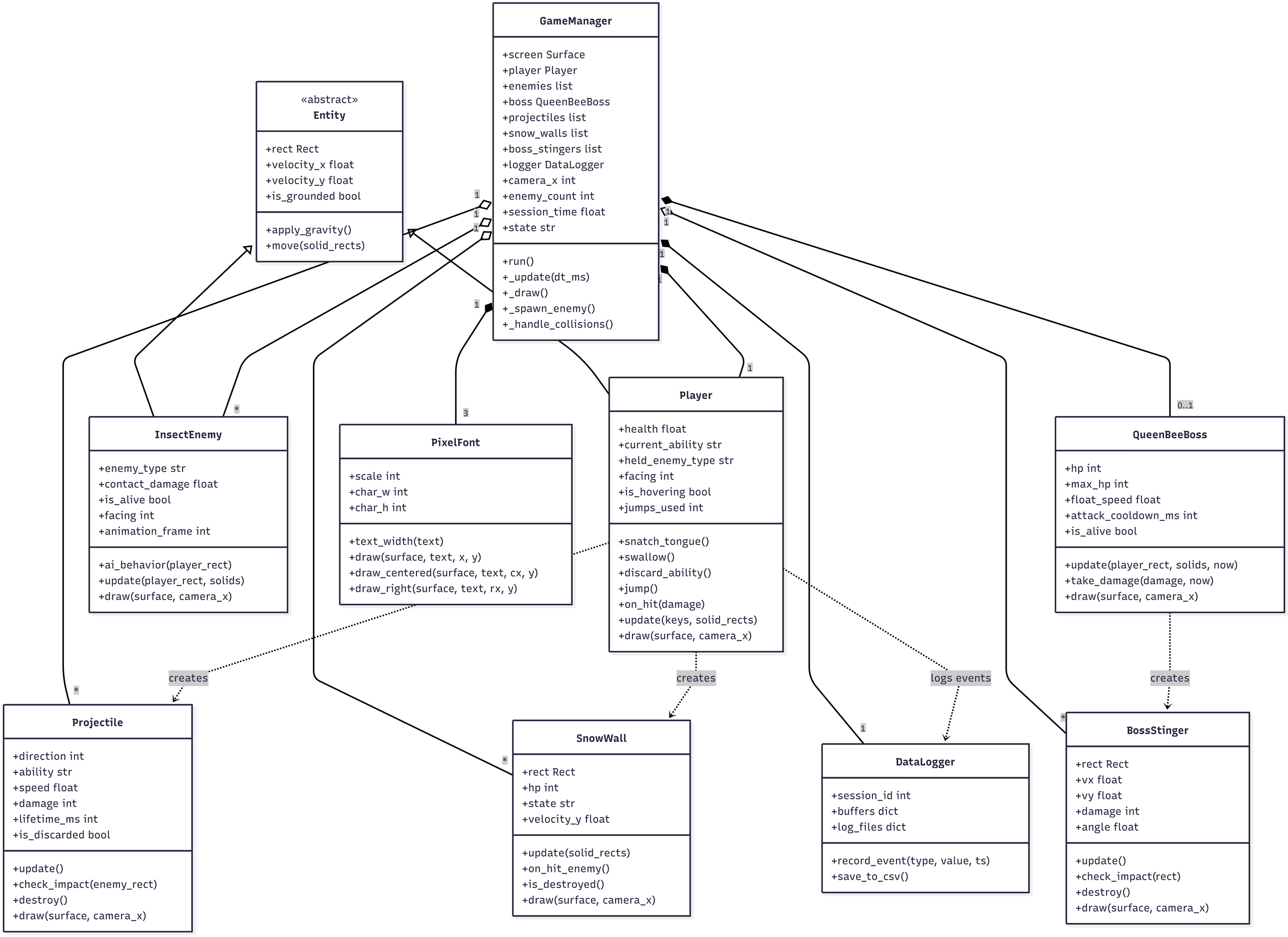
GameManager
+screen Surface
+player Player
+enemies list
+boss QueenBeeBoss
+projectiles list
+snow_walls list
+boss_stingers list
+logger DataLogger
+camera_x int
+enemy_count int
+session_time float
+state str
+run()
+_update(dt_ms)
+_draw()
+_spawn_enemy()
+_handle_collisions()
«abstract»
Entity
+rect Rect
+velocity_x float
+velocity_y float
+is_grounded bool
+apply_gravity()
+move(solid_rects)
InsectEnemy
+enemy_type str
+contact_damage float
+is_alive bool
+facing int
+animation_frame int
+ai_behavior(player_rect)
+update(player_rect, solids)
+draw(surface, camera_x)
PixelFont
+scale int
+char_w int
+char_h int
+text_width(text)
+draw(surface, text, x, y)
+draw_centered(surface, text, cx, y)
+draw_right(surface, text, rx, y)
Player
+health float
+current_ability str
+held_enemy_type str
+facing int
+is_hovering bool
+jumps_used int
+snatch_tongue()
+swallow()
+discard_ability()
+jump()
+on_hit(damage)
+update(keys, solid_rects)
+draw(surface, camera_x)
QueenBeeBoss
+hp int
+max_hp int
+float_speed float
+attack_cooldown_ms int
+is_alive bool
+update(player_rect, solids, now)
+take_damage(damage, now)
+draw(surface, camera_x)
creates
creates
logs events
creates
Projectile
+direction int
+ability str
+speed float
+damage int
+lifetime_ms int
+is_discarded bool
+update()
+check_impact(enemy_rect)
+destroy()
+draw(surface, camera_x)
SnowWall
+rect Rect
+hp int
+state str
+velocity_y float
+update(solid_rects)
+on_hit_enemy()
+is_destroyed()
+draw(surface, camera_x)
DataLogger
+session_id int
+buffers dict
+log_files dict
+record_event(type, value, ts)
+save_to_csv()
BossStinger
+rect Rect
+vx float
+vy float
+damage int
+angle float
+update()
+check_impact(rect)
+destroy()
+draw(surface, camera_x)
1
1
1
1
1
1
1
1
*
3
0..1
*
*
1
*